Reinforcement Learning: Tutorial 11

Policy gradient methods: PGT, DPG & evaluation

Week 7
University of Amsterdam

Alejandro Garcia
March 2026

Outline

1. Policy gradient methods exercises

Tutorial 11 Overview

1. Policy gradient methods exercises
2. Ask anything about HW5

Tutorial 11 Overview

1. Policy gradient methods exercises
   - Questions 10.1-10.2
2. Ask anything about HW5

Q 10.1 *Exam Question: Actor-critic algorithm

One-step Actor–Critic (episodic), for estimating $\pi_{\boldsymbol{\theta}} \approx \pi_*$

Input: a differentiable policy parameterization $\pi(a|s,\boldsymbol{\theta})$
Input: a differentiable state-value function parameterization $\hat{v}(s,\mathbf{w})$
Parameters: step sizes $\alpha^{\boldsymbol{\theta}} > 0$, $\alpha^{\mathbf{w}} > 0$
Initialize policy parameter $\boldsymbol{\theta} \in \mathbb{R}^{d'}$ and state-value weights $\mathbf{w} \in \mathbb{R}^{d}$ (e.g., to $\mathbf{0}$)
Loop forever (for each episode):
  Initialize S (first state of episode)
  $I \leftarrow 1$
  Loop while S is not terminal (for each time step):
    $A \sim \pi(\cdot|S,\boldsymbol{\theta})$
    Take action A, observe S', R
    $\delta \leftarrow R + \gamma\hat{v}(S',\mathbf{w}) - \hat{v}(S,\mathbf{w})$ (if S' is terminal, then $\hat{v}(S',\mathbf{w}) \doteq 0$)
    $\mathbf{w} \leftarrow \mathbf{w} + \alpha^{\mathbf{w}}\delta\nabla\hat{v}(S,\mathbf{w})$
    $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} + \alpha^{\boldsymbol{\theta}} I\delta\nabla\ln\pi(A|S,\boldsymbol{\theta})$
    $I \leftarrow \gamma I$
    $S \leftarrow S'$

1. In the lecture, we have used a slightly different actor-critic update, namely:

$$\theta_{t+1} = \theta_t + \alpha\ \hat{q}_{\mathbf{w}}(s_t, a_t)\nabla_\theta \log \pi_\theta(a_t|s_t)$$

There are two main differences between that update and the update in the algorithm in the figure above (other than the discount factor). Describe the two differences, and for each difference, say why the authors might have chosen to perform the update this way.

Q 10.1 *Exam Question: Actor-critic algorithm

One-step Actor–Critic (episodic), for estimating $\pi_{\boldsymbol{\theta}} \approx \pi_*$

Input: a differentiable policy parameterization $\pi(a|s, \boldsymbol{\theta})$
Input: a differentiable state-value function parameterization $\hat{v}(s, \mathbf{w})$
Parameters: step sizes $\alpha^{\boldsymbol{\theta}} > 0$, $\alpha^{\mathbf{w}} > 0$
Initialize policy parameter $\boldsymbol{\theta} \in \mathbb{R}^{d'}$ and state-value weights $\mathbf{w} \in \mathbb{R}^d$ (e.g., to $\mathbf{0}$)
Loop forever (for each episode):
  Initialize S (first state of episode)
  $I \leftarrow 1$
  Loop while S is not terminal (for each time step):
    $A \sim \pi(\cdot|S, \boldsymbol{\theta})$
    Take action A, observe S', R
    $\delta \leftarrow R + \gamma \hat{v}(S', \mathbf{w}) - \hat{v}(S, \mathbf{w})$ (if S' is terminal, then $\hat{v}(S', \mathbf{w}) \doteq 0$)
    $\mathbf{w} \leftarrow \mathbf{w} + \alpha^{\mathbf{w}} \delta \nabla \hat{v}(S, \mathbf{w})$
    $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} + \alpha^{\boldsymbol{\theta}} I \delta \nabla \ln \pi(A|S, \boldsymbol{\theta})$
    $I \leftarrow \gamma I$
    $S \leftarrow S'$

a) Instead of $q(s, a)$, δ is used. This relies on v instead of q. The reason can be that v is easier to learn than q (as it is a function of just one variable).
b) δ also includes a baseline. The baseline reduces variance just as in REINFORCE.

Q 10.1 *Exam Question: Actor-critic algorithm

One-step Actor–Critic (episodic), for estimating $\pi_{\boldsymbol{\theta}} \approx \pi_*$

Input: a differentiable policy parameterization $\pi(a|s, \boldsymbol{\theta})$
Input: a differentiable state-value function parameterization $\hat{v}(s, \mathbf{w})$
Parameters: step sizes $\alpha^{\boldsymbol{\theta}} > 0$, $\alpha^{\mathbf{w}} > 0$
Initialize policy parameter $\boldsymbol{\theta} \in \mathbb{R}^{d'}$ and state-value weights $\mathbf{w} \in \mathbb{R}^{d}$ (e.g., to $\mathbf{0}$)
Loop forever (for each episode):
  Initialize S (first state of episode)
  $I \leftarrow 1$
  Loop while S is not terminal (for each time step):
    $A \sim \pi(\cdot|S, \boldsymbol{\theta})$
    Take action A, observe S', R
    $\delta \leftarrow R + \gamma \hat{v}(S', \mathbf{w}) - \hat{v}(S, \mathbf{w})$ (if S' is terminal, then $\hat{v}(S', \mathbf{w}) \doteq 0$)
    $\mathbf{w} \leftarrow \mathbf{w} + \alpha^{\mathbf{w}} \delta \nabla \hat{v}(S, \mathbf{w})$
    $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} + \alpha^{\boldsymbol{\theta}} I \delta \nabla \ln \pi(A|S, \boldsymbol{\theta})$
    $I \leftarrow \gamma I$
    $S \leftarrow S'$

2. Actor-critic methods tend to have lower variance than actor-only methods such as REINFORCE, but can introduce a bias. Why do actor-critic methods tend to have a lower variance? And where does the additional bias come from? Briefly explain your answer.

Q 10.1 *Exam Question: Actor-critic algorithm

2. Actor-critic methods tend to have lower variance than actor-only methods such as REINFORCE, but can introduce a bias. Why do actor-critic methods tend to have a lower variance? And where does the additional bias come from? Briefly explain your answer.

   a) They have a lower variance since they do not use the (Monte-Carlo) return, which has a high variance.
   b) The use of a value function can introduce a bias, because it hasn't converged yet (initialization bias), bias during the training procedure (maximization bias), or because of the functional form of the value function (underfitting).

Q 10.2 *Exam Question: Policy gradient methods

For each of the following equations, determine whether they are unbiased estimators of the gradient $\nabla_\theta \mathbb{E}_\tau [G(\tau)]$ of undiscounted returns $G(\tau)$. Briefly explain your answer.

Q 10.2 *Exam Question: Policy gradient methods

For each of the following equations, determine whether they are unbiased estimators of the gradient $\nabla_\theta \mathbb{E}_\tau [G(\tau)]$ of undiscounted returns $G(\tau)$. Briefly explain your answer.

1. $$\mathbb{E}_\tau \left[\sum_{t'=1}^{T} r_{t'} \sum_{t=0}^{T-1} \nabla_\theta \log \pi_\theta(a_t|s_t) \right]$$

Q 10.2 *Exam Question: Policy gradient methods

For each of the following equations, determine whether they are unbiased estimators of the gradient $\nabla_\theta \mathbb{E}_\tau [G(\tau)]$ of undiscounted returns $G(\tau)$. Briefly explain your answer.

1. $$\mathbb{E}_\tau \left[\sum_{t'=1}^{T} r_{t'} \sum_{t=0}^{T-1} \nabla_\theta \log \pi_\theta(a_t|s_t) \right]$$

   Unbiased, note that this is the normal REINFORCE update.

Q 10.2 *Exam Question: Policy gradient methods

For each of the following equations, determine whether they are unbiased estimators of the gradient $\nabla_\theta \mathbb{E}_\tau [G(\tau)]$ of undiscounted returns $G(\tau)$. Briefly explain your answer.

2

$$\mathbb{E}_\tau \left[\sum_{t'=1}^{T} r_{t'} \sum_{t=t'}^{T-1} \nabla_\theta \log \pi_\theta(a_t|s_t) \right]$$

Q 10.2 *Exam Question: Policy gradient methods

For each of the following equations, determine whether they are unbiased estimators of the gradient $\nabla_\theta \mathbb{E}_\tau [G(\tau)]$ of undiscounted returns $G(\tau)$. Briefly explain your answer.

2

$$\mathbb{E}_\tau \left[\sum_{t'=1}^{T} r_{t'} \sum_{t=t'}^{T-1} \nabla_\theta \log \pi_\theta(a_t|s_t) \right]$$

These are the terms we leave out in REINFORCE v2 (the a's after the corresponding r), which are expectation 0. So we only look at the noise and we leave out the signal. Thus, it is biased (and completely useless) as estimator for the policy gradient.

Q 10.2 *Exam Question: Policy gradient methods

For each of the following equations, determine whether they are unbiased estimators of the gradient $\nabla_\theta \mathbb{E}_\tau [G(\tau)]$ of undiscounted returns $G(\tau)$. Briefly explain your answer.

3

$$\mathbb{E}_\tau \left[\sum_{t=0}^{T-1} \nabla_\theta \log \pi_\theta(a_t|s_t) \left(G_t(\tau) - \hat{V}_w(s_t) \right) \right]$$

Q 10.2 *Exam Question: Policy gradient methods

For each of the following equations, determine whether they are unbiased estimators of the gradient $\nabla_\theta \mathbb{E}_\tau [G(\tau)]$ of undiscounted returns $G(\tau)$. Briefly explain your answer.

3

$$\mathbb{E}_\tau \left[\sum_{t=0}^{T-1} \nabla_\theta \log \pi_\theta(a_t|s_t) \left(G_t(\tau) - \hat{V}_w(s_t) \right) \right]$$

Unbiased, this is REINFORCE with baseline.

Q 10.2 *Exam Question: Policy gradient methods

For each of the following equations, determine whether they are unbiased estimators of the gradient $\nabla_\theta \mathbb{E}_\tau [G(\tau)]$ of undiscounted returns $G(\tau)$. Briefly explain your answer.

4

$$\mathbb{E}_\tau \left[\sum_{t=0}^{T-1} \nabla_\theta \log \pi_\theta(a_t|s_t) q_\pi(a_t, s_t) \right]$$

Q 10.2 *Exam Question: Policy gradient methods

For each of the following equations, determine whether they are unbiased estimators of the gradient $\nabla_\theta \mathbb{E}_\tau [G(\tau)]$ of undiscounted returns $G(\tau)$. Briefly explain your answer.

4.

$$\mathbb{E}_\tau \left[\sum_{t=0}^{T-1} \nabla_\theta \log \pi_\theta(a_t|s_t) q_\pi(a_t, s_t) \right]$$

Unbiased (return is true q_π, which gives the correct target for the policy update).

Q 10.2 *Exam Question: Policy gradient methods

For each of the following equations, determine whether they are unbiased estimators of the gradient $\nabla_\theta \mathbb{E}_\tau [G(\tau)]$ of undiscounted returns $G(\tau)$. Briefly explain your answer.

5

$$\mathbb{E}_\tau \left[\sum_{t=0}^{T-1} \nabla_\theta \log \pi_\theta(a_t|s_t) \left(G_t(\tau) - \hat{q}_w(s_t, a_t) \right) \right]$$

Q 10.2 *Exam Question: Policy gradient methods

For each of the following equations, determine whether they are unbiased estimators of the gradient $\nabla_\theta \mathbb{E}_\tau [G(\tau)]$ of undiscounted returns $G(\tau)$. Briefly explain your answer.

5

$$\mathbb{E}_\tau \left[\sum_{t=0}^{T-1} \nabla_\theta \log \pi_\theta(a_t|s_t) \left(G_t(\tau) - \hat{q}_w(s_t, a_t) \right) \right]$$

Biased, baseline depends on a_t which depends on trajectory.

Q 10.2 *Exam Question: Policy gradient methods

For each of the following equations, determine whether they are unbiased estimators of the gradient $\nabla_\theta \mathbb{E}_\tau [G(\tau)]$ of undiscounted returns $G(\tau)$. Briefly explain your answer.

6

$$\mathbb{E}_\tau \left[\sum_{t=0}^{T-1} \nabla_\theta \log \pi_\theta(a_t|s_t) \hat{q}_w(s_t, a_t) \right]$$

Q 10.2 *Exam Question: Policy gradient methods

For each of the following equations, determine whether they are unbiased estimators of the gradient $\nabla_\theta \mathbb{E}_\tau [G(\tau)]$ of undiscounted returns $G(\tau)$. Briefly explain your answer.

6

$$\mathbb{E}_\tau \left[\sum_{t=0}^{T-1} \nabla_\theta \log \pi_\theta(a_t|s_t) \hat{q}_w(s_t, a_t) \right]$$

Biased since q_w is an approximation to q_π and might not be exact. (In an optional module we covered the compatible function approximation theorem which tells us that the estimator is only unbiased if the value function is compatible s.t. expected value remains unchanged.)